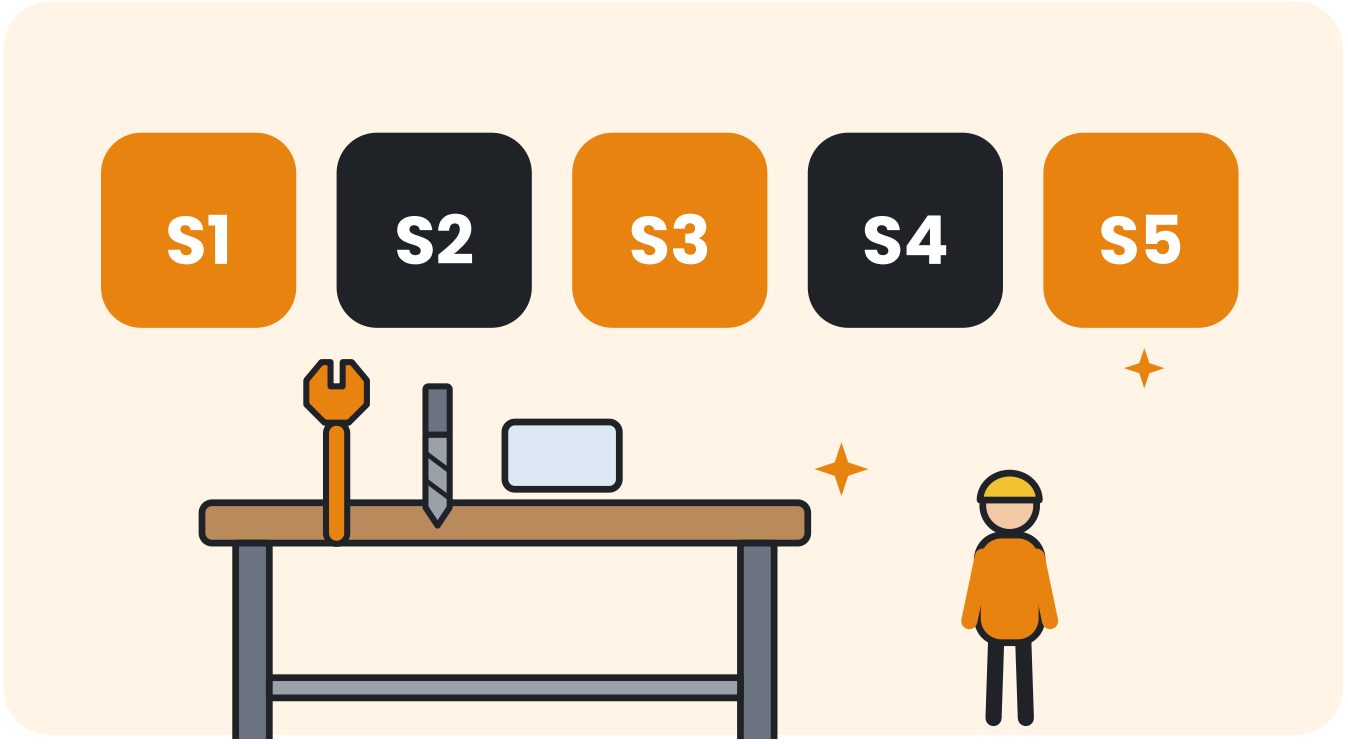
MachiNest

LEAN MANUFACTURING BASICS

5S: A Clean, Organised and Safe Workplace

A simple guide for CNC and machine shops, with shop-floor examples for each step.

Sort | Set in order | Shine | Standardize | Sustain



INTRODUCTION

What is 5S?

5S is a method from Japan for organising the workplace. It removes what is not needed, gives everything a fixed place, keeps the area clean, and turns these habits into a daily routine.

Why it matters on the shop floor

- Less time searching for tools, gauges and drawings
- Faster setups and fewer machine stoppages
- Fewer mix-ups between OK, rework and reject parts
- Problems like leaks and loose parts are seen early
- A safer floor: fewer slips, trips and cuts
- Customers and auditors see a well-run shop

#	Japanese	English	Hindi	In one line
1	Seiri	**Sort**	छँटाई	Keep only what is needed
2	Seiton	**Set in order**	सुव्यवस्थित करना	A place for everything
3	Seiso	**Shine**	सफ़ाई	Clean and inspect
4	Seiketsu	**Standardize**	मानकीकरण	Same rules everywhere
5	Shitsuke	**Sustain**	अनुशासन	Make it a habit

Your first 5S week

Day 1
Pick one area, form a small team and take photos.

Day 2
Sort: red-tag everything not needed.

Day 3
Set in order: shadow board, labels, locations.

Day 4
Shine: deep clean, find and tag leaks.

Day 5
Standardize: floor marking and the first audit.

After that, keep the weekly audit going. That is Sustain.

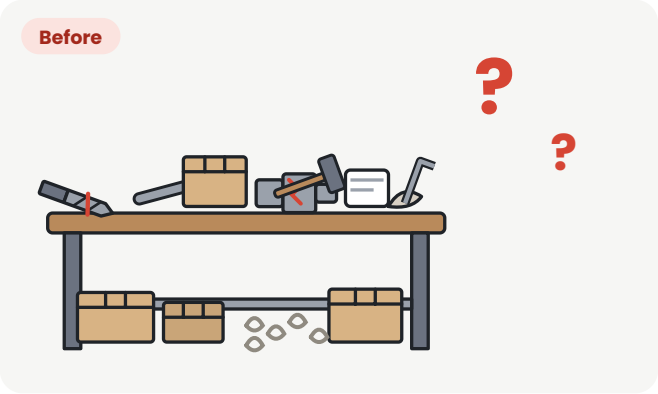
Start small Pick one machine or one workbench. Finish all five steps there, take before and after photos, and then move to the next area.

S1 · SEIRI

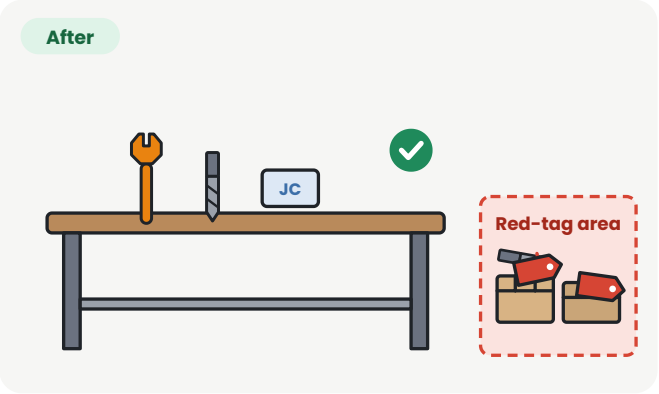
1 Sort

Separate what is needed from what is not, and remove everything that is not needed for today's work.

Key question "Do I need this item here, for the work we do now?"



Broken tools, old drawings, scrap and unused boxes hide the items that matter.



Only needed items stay at the bench. Doubtful items go to the red-tag area for a decision.

How to do it

1. Take photos of the area as it is now.
2. Check every item: tools, gauges, holders, drawings, boxes and parts.
3. Put a red tag on anything not needed, broken or of unknown use.
4. Move red-tagged items to one marked red-tag area.
5. Within 1–2 weeks, decide: use elsewhere, repair, sell or scrap.

Shop-floor examples

How often it is used	Where to keep it
Every cycle or every hour	At the machine
Every day	In the cell or a nearby rack
Every week or month	In the tool store
Rarely or never	Red tag: decide and remove

Common mistake

Keeping items "just in case". If it has not been used in the last month, it does not belong at the machine.

Shop tip

Worn inserts, broken drills and old part samples are the most common items left at CNC machines. Clear them first.

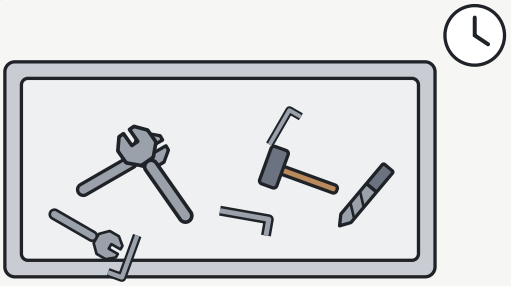
S2 · SEITON

2 Set in order

Give every needed item a fixed, marked place so anyone can find it, use it and put it back quickly.

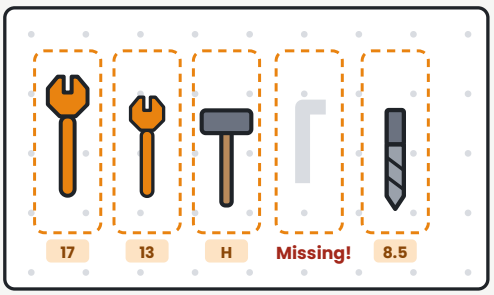
Key question "Can a new operator find this item in 30 seconds?"

Before



Tools mixed in a drawer. Time is lost searching and tools get damaged.

After



Shadow board with outlines and size labels. A missing tool is seen at once.

How to do it

1. Decide the best place for each item, closest to where it is used.
2. Make a shadow board, foam cut-out or labelled rack.
3. Label each place with name, size or tool number.
4. Keep the most used items between knee and shoulder height.
5. Mark locations for bins, trolleys and gauges on the floor or shelf.

Shop-floor examples

- Spanners and Allen keys on a shadow board next to the machine
- Tool holders in a rack labelled by T-number (T0101, T0202...)
- Inserts sorted by code and grade in labelled boxes
- Plug and thread gauges in a foam tray with cut-outs
- Drawings and inspection reports in a holder at each machine

Common mistake

Labelling once and never updating. When a tool changes, update the outline and label the same day.

Rule

"A place for everything, and everything in its place." If an item has no marked place, it does not have a place yet.

S3 · SEISO

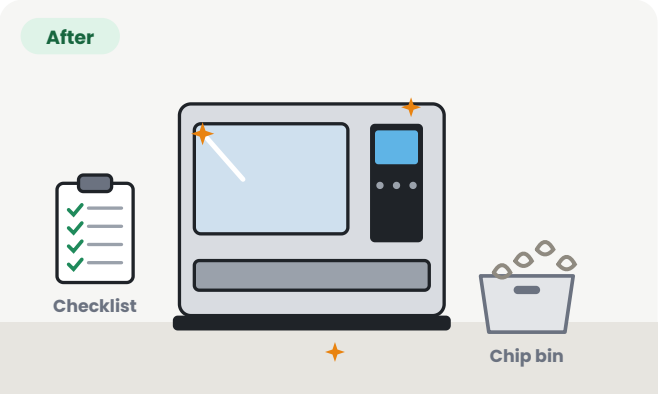
3 Shine

Clean the machine and the area every day, and use cleaning as a chance to inspect the machine.

Key question "While cleaning, what problem can I see before it becomes a breakdown?"



Chips on the floor, oil on the floor and a dirty window. Leaks and damage stay hidden.



Clean machine, chip bin in place and a daily checklist filled by the operator.

How to do it

1. Divide the area and give each person a clear cleaning zone.
2. Clean at a fixed time, for example the last 10 minutes of the shift.
3. Look for leaks, loose bolts, damaged covers and abnormal noise while cleaning.
4. Tag each problem found and report it to maintenance.
5. Fix the source of dirt where possible (chip guards, drip trays).

Shop-floor examples

Daily check (example)	Time
Remove chips from table, chuck and way covers	3 min
Wipe the window, panel and door handle	2 min
Check coolant and lubrication oil level	1 min
Clean floor around the machine, empty chip bin	3 min
Look for leaks, loose parts and damage	1 min

Common mistake

Cleaning only before an audit or a customer visit. Clean every day, at a fixed time.

Remember

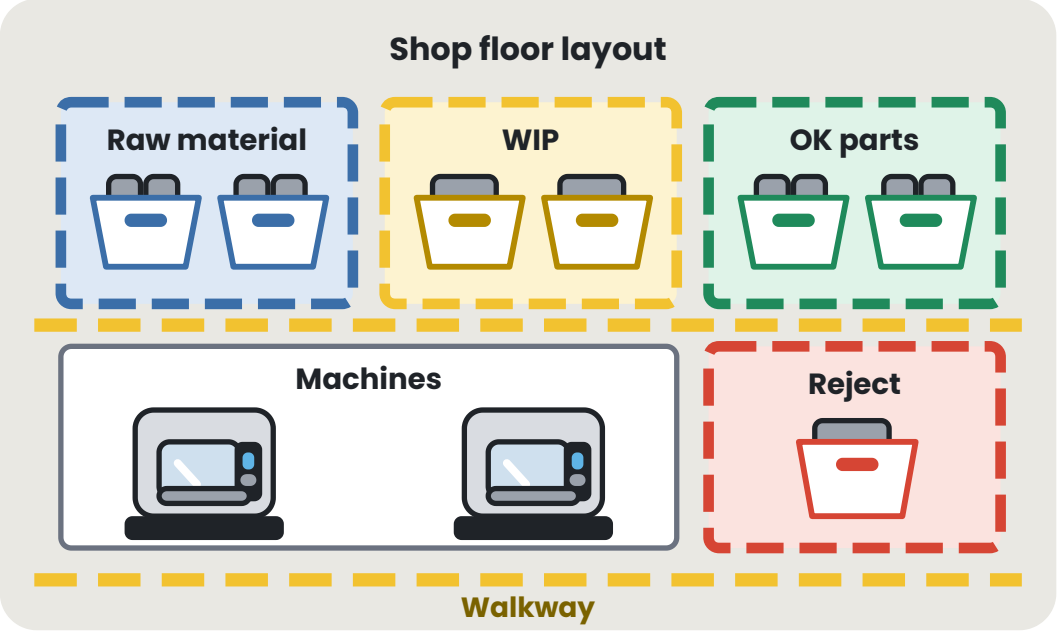
Cleaning is inspection. A small oil leak found today is much cheaper than a spindle or guideway repair later.

S4 · SEIKETSU

4 Standardize

Make the first three S's the same everywhere, with simple visual rules that everyone follows.

Key question "Can anyone see at a glance whether this area is normal or not?"



Example floor layout: colour-coded zones keep raw material, WIP, OK and reject parts apart, with clear walkways.

How to do it

1. Choose one colour code for the whole shop and write it down.
2. Mark floor zones and label every rack, bin and location.
3. Put a photo of the "standard condition" at each machine.
4. Make simple checklists and one-point lessons for each area.
5. Add 5S tasks to daily work, not as an extra job.

Shop-floor examples

Colour (example)	Use
Yellow	Walkways and aisle lines
Green	OK and finished parts
Red	Reject, scrap, fire equipment
Blue	Raw material
Yellow-black stripes	Hazard areas

Common mistake
Each area using its own colours and rules. One shop, one standard.

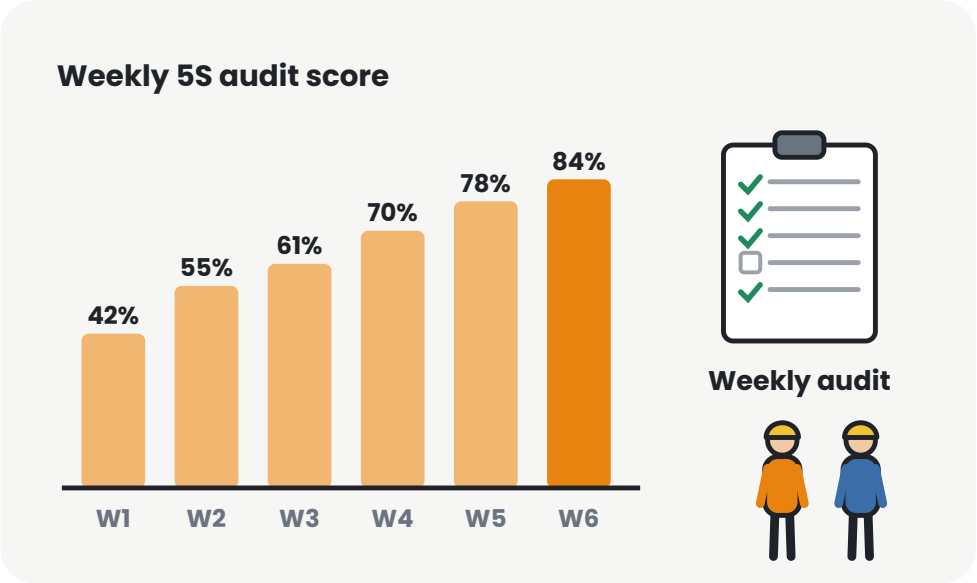
Shop tip
A reject bin that is red and kept in a red zone prevents bad parts from being packed with good parts.

S5 · SHITSUKE

5 Sustain

Keep doing 5S every day until it becomes a habit, and check it regularly so the shop does not slide back.

Key question "Are we still following the standard we agreed on?"



Display the weekly audit score where everyone can see it. A rising score shows the habit is forming.

How to do it

1. Do a short 5S audit every week in each area (see the checklist on the next page).
2. Display the score and photos on a notice board.
3. Close the actions found in the audit before the next audit.
4. Supervisors and managers walk the floor and ask about 5S.
5. Train new joiners on 5S on their first day.
6. Appreciate the best area every month.

Daily habits

- Put the tool back right after use
- Clean the machine before handing over the shift
- Report any abnormality the same day
- Do not keep personal items at the machine

Common mistake

Only managers doing the audit. Let operators audit each other's areas too.

Remember

5S is not a one-time cleaning drive. Without S5, most areas slide back within a few weeks.

PRINTABLE CHECKLIST

Weekly 5S audit checklist

Score each point: 0 = not done, 1 = poor, 2 = average, 3 = good, 4 = excellent. Maximum score = 60.

Area / machine _____ Date _____ Audited by _____

S	Check point	Score (0–4)
Sort	Only needed tools, gauges and parts are at the machine	
Sort	No broken tools, old drawings or scrap lying around	
Sort	Red-tag area exists and is cleared every week	
Set in order	Every tool and gauge has a marked place	
Set in order	Tool holders and inserts are labelled and found within 30 seconds	
Set in order	Bins, trolleys and material are in marked locations	
Shine	Machine, window and way covers are free of chips and oil	
Shine	Floor is clean, with no coolant or oil leaks	
Shine	Daily cleaning checklist is filled and signed	
Standardize	Floor marking and colour code are followed	
Standardize	Checklists and one-point lessons are displayed	
Standardize	OK, WIP and reject parts are in their own marked zones	
Sustain	Last audit score is displayed on the board	
Sustain	Actions from the last audit are closed	
Sustain	Everyone in the area knows the 5S rules	

Total score (out of 60)

_____

Score % = total ÷ 60 × 100

Top 3 actions for next week

_____

_____

_____